# PHP-FL class fairness

Complete results compendium

All committed result tables, including A/B/C, the five half-scale runs, the FedAPEN scope test, and the four-step diagnostic ladder.

Prepared 10 November 2026

Repository: github.com/Paul4805/phofl-class-fairness

# Experiment inventory and status

| Experiment | Status | Key result |
|---|---|---|
| Original Table-2 baseline (AM 66.94 / FM 8.07) | Not executed | No reproduction claim |
| A: fairness metrics (Fashion-MNIST smoke) | Complete | FM 0.1607; tail not tail-worse |
| B: aggregation reweighting (smoke) | Complete | Null (p 0.469-0.922) |
| C: correlated participation (smoke) | Complete | Pattern effect loses Holm |
| C pinned-stack smoke | Complete | Signs replicate; significance does not |
| C half-scale (4 runs) | Complete | FAIL pre-registered escalation |
| FedAPEN / linear half-scale | Complete | 52.9-point gap; field-wide scope |
| Ladder 1-4 (half-scale) | Complete | Max 7.9% gap reduction |
| Seed-1 rerun | Not triggered | No step reached 30% |
| FedMRL / FedFV / FedHEAL / FedAWE / FedAU | Not run | Out of scope |

All half-scale numbers use 10 clients, 60 rounds, one seed, four CNNs, one local epoch, 2,500 train / 500 test per client, and the pinned stack. Absolute AM/FM are not comparable to the paper's full-scale numbers.

# A/B/C: Fashion-MNIST smoke results

## A. Fairness metrics across completed smoke runs

| exp | method | pattern | agg_mode | AM | FM | min_acc | p10 | cvar10 | jain | rho_p_acc |
|---|---|---|---|---|---|---|---|---|---|---|
| B | PHPFL | normal | inv_N | 0.5105 | 0.1615 | 0.2600 | 0.2915 | 0.2600 | 0.9090 | 0.0667 |
| B | PHPFL | normal | inv_p | 0.5160 | 0.1628 | 0.2600 | 0.2780 | 0.2600 | 0.9095 | -0.0182 |
| B | PHPFL | normal | plain | 0.5175 | 0.1607 | 0.2400 | 0.2985 | 0.2400 | 0.9120 | -0.0061 |
| C | PHPFL | entropy | plain | 0.5120 | 0.1805 | 0.1750 | 0.2425 | 0.1750 | 0.8894 | 0.7212 |
| C | PHPFL | linear | plain | 0.5040 | 0.1439 | 0.2650 | 0.2875 | 0.2650 | 0.9246 | -0.7576 |
| C | Standalone | linear | none | 0.4605 | 0.1856 | 0.1000 | 0.1990 | 0.1000 | 0.8603 | -0.7939 |

A's original narrative (lowest FM but not best tail) was not confirmed: the FM winner also won min/CVaR, and rho was not positive.

## B. Aggregation reweighting

| agg_mode | AM | FM | min |
|---|---|---|---|
| plain | 0.5175 | 0.1607 | 0.2400 |
| inv_p | 0.5160 | 0.1628 | 0.2600 |
| inv_N | 0.5105 | 0.1615 | 0.2600 |

| comparison | statistic | p | median diff |
|---|---|---|---|
| plain vs inv_p [accuracy] | 26.0000 | 0.9219 | -0.0125 |
| plain vs inv_N [accuracy] | 16.0000 | 0.4688 | 0.0050 |
| inv_p vs inv_N [accuracy] | 17.0000 | 0.5703 | 0.0050 |

## C. Correlated participation

| pattern | method | AM | FM | min |
|---|---|---|---|---|
| entropy_correlated | PHPFL | 0.5120 | 0.1805 | 0.1750 |
| entropy_correlated | Standalone | 0.4605 | 0.1856 | 0.1000 |
| linear | PHPFL | 0.5040 | 0.1439 | 0.2650 |
| linear | Standalone | 0.4605 | 0.1856 | 0.1000 |

| C comparison | raw p | Holm p | loses sig |
|---|---|---|---|
| PHPFL vs Standalone [linear] | 0.0645 | 0.1934 | False |
| PHPFL vs Standalone [entropy_correlated] | 0.0020 | 0.0098 | False |
| linear vs entropy [accuracy] | 0.4102 | 0.8203 | False |
| linear vs entropy [dispersion] | 0.0410 | 0.1641 | True |
| linear vs entropy [advantage] | 0.4102 | 0.8203 | False |

The dispersion effect lost Holm significance; PHP-FL vs Standalone under entropy survived. C is closed.

# C escalation: pre-registered thresholds and outcomes

| Quantity | Value | Rule |
|---|---|---|
| FM increase | -2.84% | GO over 25%; STOP under 10% |
| Min-accuracy change | -4.09 pts | GO over 5 drop; STOP under 2 drop |
| Rho difference | 0.1697 | GO over 1.0 |
| Effect vs pinned smoke | -2.84% vs 15.42% | STOP if at most 1.1x pinned |
| Decision | FAIL | STOP / GO / MARGINAL |

## Pinned-stack smoke C

| pattern | method | AM | FM | min | p10 | CVaR10 | rho |
|---|---|---|---|---|---|---|---|
| linear | PHPFL | 0.5155 | 0.1511 | 0.2450 | 0.2990 | 0.2450 | -0.7212 |
| linear | Standalone | 0.4575 | 0.1826 | 0.1050 | 0.1950 | 0.1050 | -0.8424 |
| entropy_correlated | PHPFL | 0.5120 | 0.1744 | 0.2300 | 0.2570 | 0.2300 | 0.6930 |
| entropy_correlated | Standalone | 0.4575 | 0.1826 | 0.1050 | 0.1950 | 0.1050 | 0.6727 |

| pinned comparison | raw p | Holm p | loses sig |
|---|---|---|---|
| PHPFL vs Standalone [linear] | 0.0195 | 0.0781 | True |
| PHPFL vs Standalone [entropy_correlated] | 0.0020 | 0.0098 | False |
| linear vs entropy [accuracy] | 0.9219 | 1.0000 | False |
| linear vs entropy [dispersion] | 0.0840 | 0.2520 | False |
| linear vs entropy [advantage] | 0.8457 | 1.0000 | False |

The pinned stack preserves the sign but not the significance of the dispersion effect. PHP-FL vs Standalone under entropy remains significant; the pattern effect does not.

# Five half-scale runs: two-axis fairness

| Run | AM | client-FM | class_std | class gap | support gap | worst class | worst recall | best class | best recall |
|---|---|---|---|---|---|---|---|---|---|
| PHPFL/lin | 0.6320 | 0.0785 | 0.1334 | 0.4430 | 0.5125 | dog | 0.2510 | frog | 0.6940 |
| PHPFL/ent | 0.6308 | 0.0763 | 0.1349 | 0.4560 | 0.4499 | dog | 0.2400 | truck | 0.6960 |
| FedAPEN/lin | 0.6209 | 0.0839 | 0.1698 | 0.5290 | 0.4159 | cat | 0.2560 | frog | 0.7850 |
| Standalone/lin | 0.5429 | 0.0962 | 0.1281 | 0.3860 | 0.3502 | cat | 0.2290 | frog | 0.6150 |
| Standalone/ent | 0.5467 | 0.0985 | 0.1196 | 0.3890 | 0.3541 | cat | 0.2180 | frog | 0.6070 |

Conservative headline: 44.3 points (PHP-FL/linear). Best client-FM run: PHP-FL/entropy at 45.6 points. FedAPEN: 52.9 points.

## Per-class recall, full CIFAR-10 test set

| run | airplane | automobile | bird | cat | deer | dog | frog | horse | ship | truck |
|---|---|---|---|---|---|---|---|---|---|---|
| PHPFL/lin | 0.609 | 0.571 | 0.610 | 0.434 | 0.382 | 0.251 | 0.694 | 0.588 | 0.623 | 0.662 |
| PHPFL/ent | 0.626 | 0.613 | 0.599 | 0.457 | 0.397 | 0.240 | 0.673 | 0.511 | 0.616 | 0.696 |
| FedAPEN/lin | 0.520 | 0.704 | 0.631 | 0.256 | 0.482 | 0.275 | 0.785 | 0.639 | 0.663 | 0.424 |
| Standalone/lin | 0.405 | 0.586 | 0.564 | 0.229 | 0.335 | 0.285 | 0.615 | 0.381 | 0.488 | 0.541 |
| Standalone/ent | 0.451 | 0.579 | 0.531 | 0.218 | 0.369 | 0.306 | 0.607 | 0.398 | 0.476 | 0.554 |

Class gap = max - min of this row. Weak region: dog/cat/deer/horse.

## Support filtering

| run | total cells | supported | low support | threshold |
|---|---|---|---|---|
| PHPFL/lin | 100 | 66 | 34 | 10 |
| PHPFL/ent | 100 | 66 | 34 | 10 |
| Standalone/lin | 100 | 66 | 34 | 10 |
| Standalone/ent | 100 | 66 | 34 | 10 |
| FedAPEN/lin | 100 | 66 | 34 | 10 |

170 low-support cells total; 45 zero-sample cells. Worst supported cell is 0.0 (client 0, cat, 11 samples).

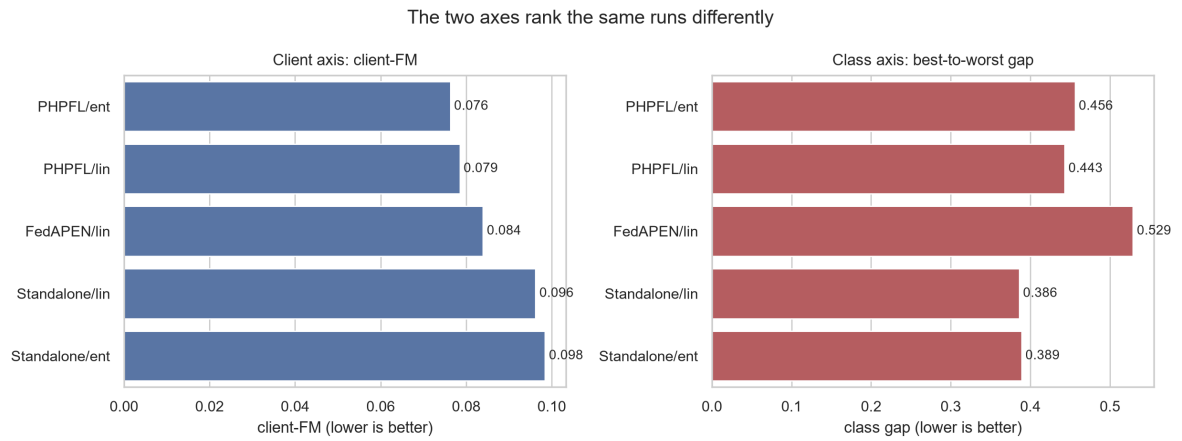
# Ranking disagreement and AM normalization

| run | client-FM | FM rank | class_std | std rank | class gap | gap rank |
|---|---|---|---|---|---|---|
| PHPFL/ent | 0.0763 | 1 | 0.1349 | 4 | 0.4560 | 4 |
| PHPFL/lin | 0.0785 | 2 | 0.1334 | 3 | 0.4430 | 3 |
| FedAPEN/lin | 0.0839 | 3 | 0.1698 | 5 | 0.5290 | 5 |
| Standalone/lin | 0.0962 | 4 | 0.1281 | 2 | 0.3860 | 1 |
| Standalone/ent | 0.0985 | 5 | 0.1196 | 1 | 0.3890 | 2 |

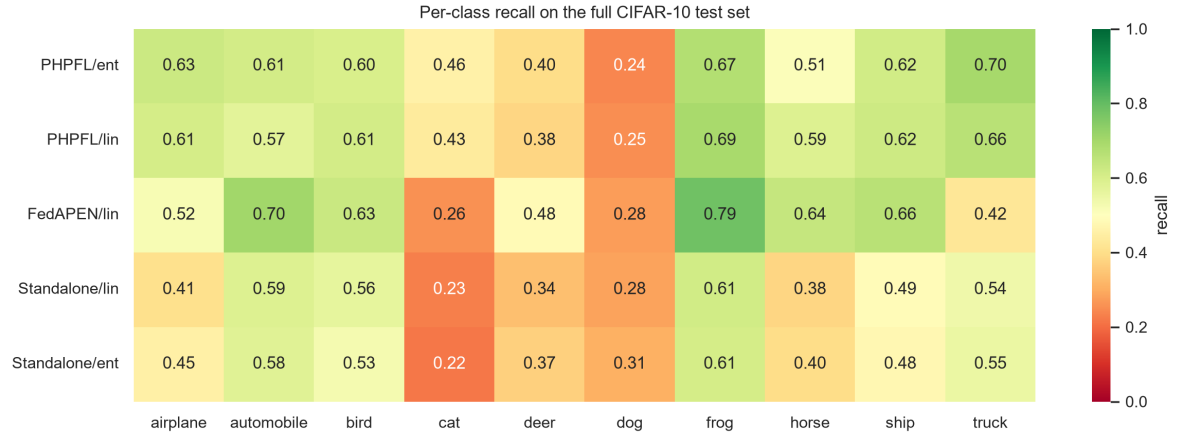
FedAPEN is 3rd by client-FM but 5th by class gap. Standalone/linear is 4th by client-FM but 1st by class gap. n=5, observation only.

| run | AM | raw gap/AM | raw gap/ceil | support gap | support gap/AM | support gap/ceil |
|---|---|---|---|---|---|---|
| PHPFL/lin | 0.6320 | 0.7010 | 0.5915 | 0.5125 | 0.8110 | 0.6803 |
| PHPFL/ent | 0.6308 | 0.7229 | 0.6000 | 0.4499 | 0.7132 | 0.6399 |
| Standalone/lin | 0.5429 | 0.7110 | 0.5006 | 0.3502 | 0.6450 | 0.4363 |
| Standalone/ent | 0.5467 | 0.7115 | 0.4974 | 0.3541 | 0.6477 | 0.4579 |
| FedAPEN/lin | 0.6209 | 0.8519 | 0.7110 | 0.4159 | 0.6697 | 0.5484 |

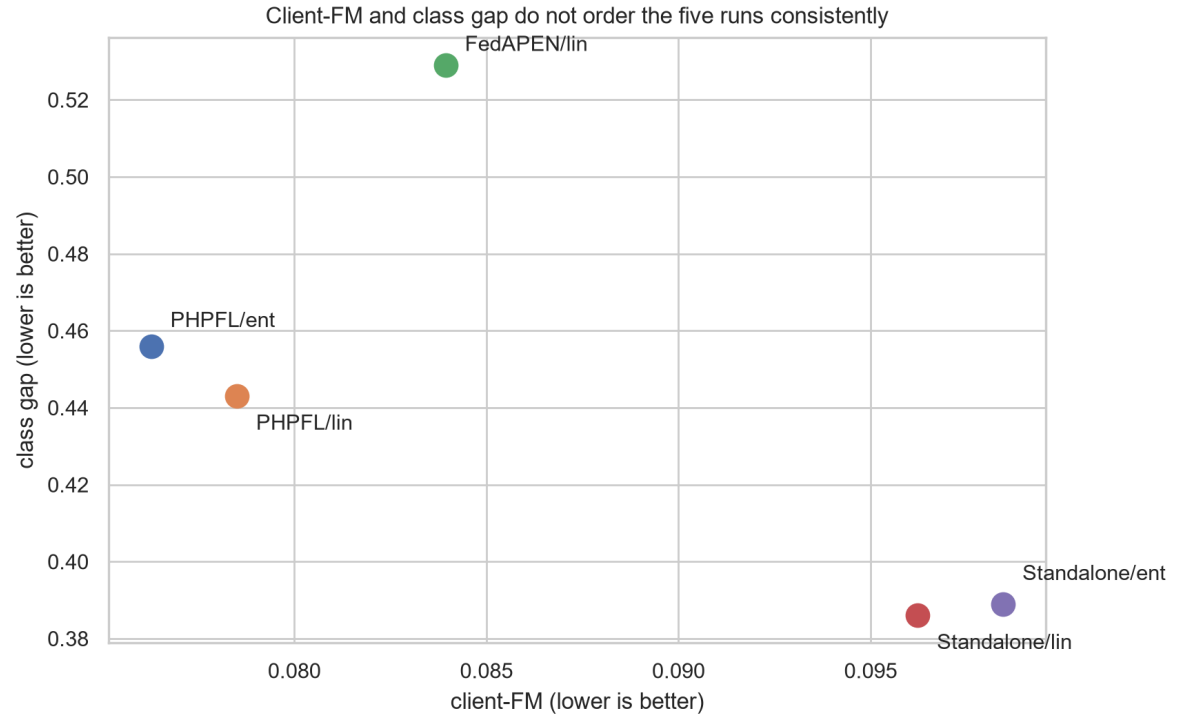
Standalone's raw gap advantage disappears after AM normalization (0.711 vs 0.701). The PHP-FL vs FedAPEN inversion survives (FedAPEN 3rd FM, 5th gap/AM).




Client-FM and class gap across the five runs.




Per-class recall matrix.

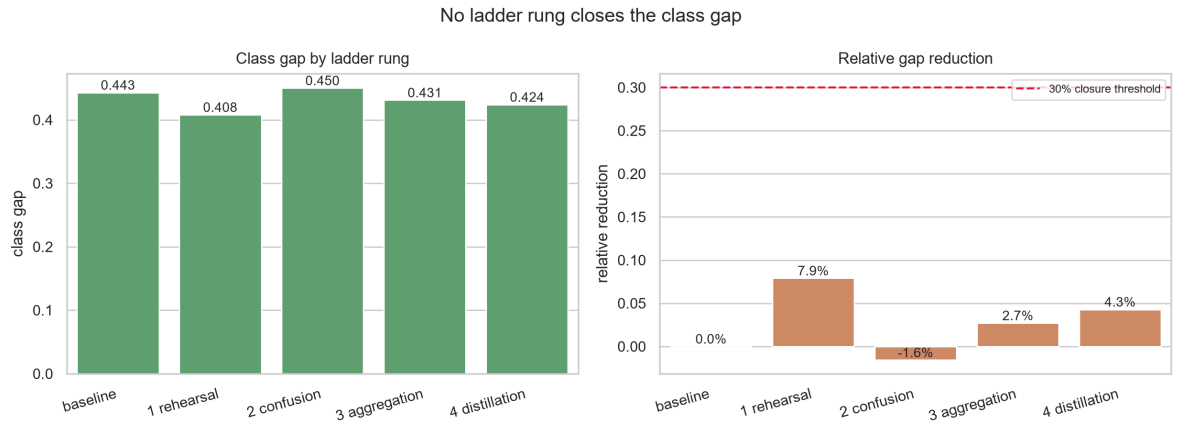



Client-FM versus class gap; the ordering is not monotone.

# Diagnostic ladder: all four interventions

| Rung | Intervention | AM | FM | gap | reduction | dog | frog | node |
|---|---|---|---|---|---|---|---|---|
| 0 | PHP-FL/linear baseline | 0.6320 | 0.0785 | 0.443 | - | 0.251 | 0.694 | - |
| 1 | class-balanced rehearsal | 0.6099 | 0.0854 | 0.4080 | +7.9% | 0.3200 | 0.7280 | dog 0.320 |
| 2 | confusion-targeted loss | 0.6346 | 0.0900 | 0.4500 | -1.6% | 0.2700 | 0.7200 | dog 0.270 |
| 3 | worst-class aggregation | 0.6312 | 0.1002 | 0.4310 | +2.7% | 0.2920 | 0.7230 | dog 0.292 |
| 4 | per-class distillation | 0.6322 | 0.0944 | 0.4240 | +4.3% | 0.2820 | 0.7060 | dog 0.282 |

No rung reaches the 30% closure threshold; maximum reduction 7.9%. No seed-1 rerun was triggered.



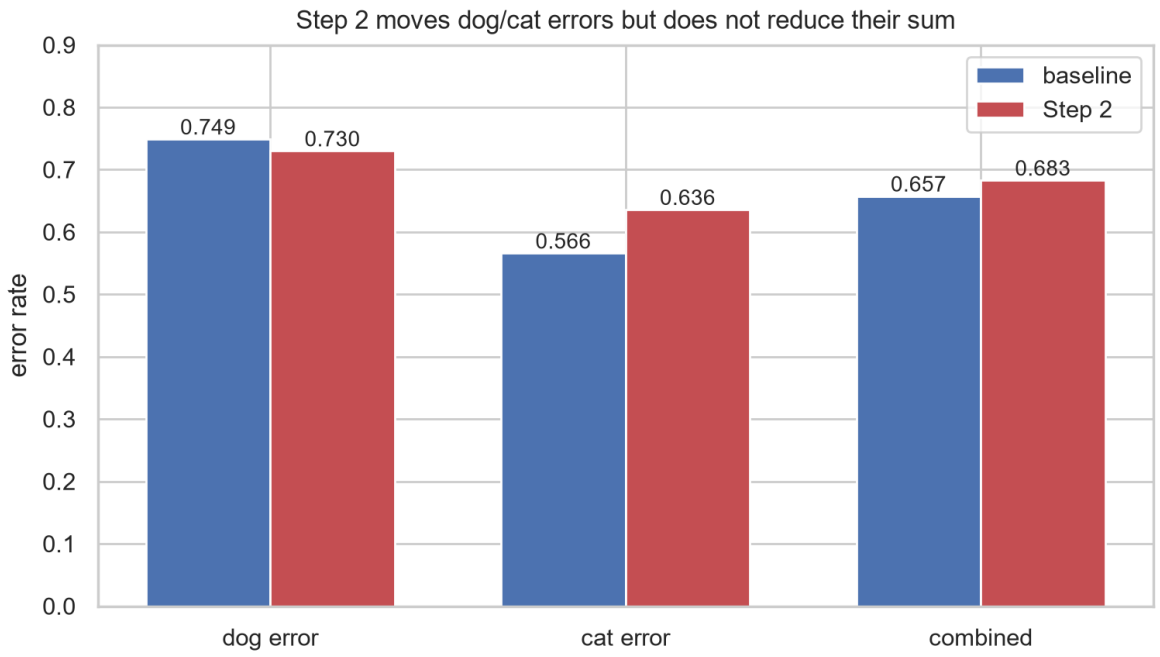

Ladder class gap and relative reduction.

| Rung | Observed detail |
|---|---|
| 1 | dog 0.251 -> 0.320; frog 0.694 -> 0.728; gap 0.443 -> 0.408 (7.9%). |
| 2 | dog->cat 0.249 -> 0.191; cat recall 0.434 -> 0.364; combined dog+cat error 0.657 -> 0.683. |
| 3 | gamma=5.0; worst class airplane early, dog later; dog recall 0.292; gap 0.431 (2.7%). |
| 4 | T=1.0; best dog architecture EfficientNet-B1 0.599 -> 0.629, unchanged; gap 0.424 (4.3%). |

## Step-by-step per-class results

| step | class | baseline | method | delta |
|---|---|---|---|---|
| 1 | airplane | +0.6090 | +0.6060 | -0.0030 |
| 1 | automobile | +0.5710 | +0.6570 | +0.0860 |
| 1 | bird | +0.6100 | +0.4820 | -0.1280 |
| 1 | cat | +0.4340 | +0.3770 | -0.0570 |
| 1 | deer | +0.3820 | +0.5040 | +0.1220 |
| 1 | dog | +0.2510 | +0.3200 | +0.0690 |
| 1 | frog | +0.6940 | +0.7280 | +0.0340 |
| 1 | horse | +0.5880 | +0.5690 | -0.0190 |
| 1 | ship | +0.6230 | +0.6330 | +0.0100 |
| 1 | truck | +0.6620 | +0.5910 | -0.0710 |
| 2 | airplane | +0.6090 | +0.5960 | -0.0130 |
| 2 | automobile | +0.5710 | +0.6050 | +0.0340 |
| 2 | bird | +0.6100 | +0.6020 | -0.0080 |
| 2 | cat | +0.4340 | +0.3640 | -0.0700 |
| 2 | deer | +0.3820 | +0.4270 | +0.0450 |
| 2 | dog | +0.2510 | +0.2700 | +0.0190 |
| 2 | frog | +0.6940 | +0.7200 | +0.0260 |
| 2 | horse | +0.5880 | +0.5670 | -0.0210 |
| 2 | ship | +0.6240 | +0.6540 | +0.0300 |
| 2 | truck | +0.6620 | +0.6840 | +0.0220 |
| 3 | airplane | +0.6090 | +0.5780 | -0.0310 |
| 3 | automobile | +0.5710 | +0.6170 | +0.0460 |
| 3 | bird | +0.6100 | +0.5600 | -0.0500 |
| 3 | cat | +0.4340 | +0.4070 | -0.0270 |
| 3 | deer | +0.3820 | +0.4420 | +0.0600 |
| 3 | dog | +0.2510 | +0.2920 | +0.0410 |
| 3 | frog | +0.6940 | +0.7230 | +0.0290 |
| 3 | horse | +0.5880 | +0.6200 | +0.0320 |
| 3 | ship | +0.6230 | +0.6580 | +0.0350 |
| 3 | truck | +0.6620 | +0.6400 | -0.0220 |
| 4 | airplane | +0.6090 | +0.5830 | -0.0260 |
| 4 | automobile | +0.5710 | +0.5670 | -0.0040 |
| 4 | bird | +0.6100 | +0.5900 | -0.0200 |
| 4 | cat | +0.4340 | +0.4570 | +0.0230 |
| 4 | deer | +0.3820 | +0.3930 | +0.0110 |
| 4 | dog | +0.2510 | +0.2820 | +0.0310 |
| 4 | frog | +0.6940 | +0.7060 | +0.0120 |
| 4 | horse | +0.5880 | +0.5790 | -0.0090 |
| 4 | ship | +0.6230 | +0.6650 | +0.0420 |
| 4 | truck | +0.6620 | +0.6660 | +0.0040 |

Step 2 zero-sum: dog->cat 0.249 -> 0.191 (-23%); cat recall 0.434 -> 0.364 (-16%); combined dog+cat error 0.657 -> 0.683 (rose).




Step 2 shifts dog/cat errors without reducing their combined rate.

# All statistical tests

## Half-scale C tests (raw and Holm)

| metric | comparison | test | n | stat | raw p | Holm p | Holm sig |
|---|---|---|---|---|---|---|---|
| accuracy | PHPFL vs Standalone [linear] | wilcoxon | 10 | 0.0000 | 0.0020 | 0.0176 | True |
| accuracy | PHPFL vs Standalone [entropy_correlated] | wilcoxon | 10 | 0.0000 | 0.0020 | 0.0176 | True |
| accuracy | linear vs entropy [accuracy] | wilcoxon | 10 | 26.5000 | 1.0000 | 1.0000 | False |
| \|acc - group mean\| | linear vs entropy [dispersion] | wilcoxon | 10 | 27.0000 | 1.0000 | 1.0000 | False |
| min-accuracy tail shortfall | linear vs entropy [min tail-shortfall] | wilcoxon | 10 | 7.0000 | 0.8927 | 1.0000 | False |
| CVaR10 tail shortfall | linear vs entropy [CVaR10 tail-shortfall] | wilcoxon | 10 | 7.0000 | 0.8927 | 1.0000 | False |
| min-accuracy | linear vs entropy [min] | paired bootstrap | 10 | -0.0256 | 0.2088 | 1.0000 | False |
| CVaR10 | linear vs entropy [CVaR10] | paired bootstrap | 10 | -0.0256 | 0.2088 | 1.0000 | False |
| PHPFL - Standalone | linear vs entropy [advantage] | wilcoxon | 10 | 26.5000 | 1.0000 | 1.0000 | False |

## B smoke tests

| metric | comparison | n | stat | p | median diff |
|---|---|---|---|---|---|
| accuracy | plain vs inv_p [accuracy] | 10 | 26.0000 | 0.9219 | -0.0125 |
| accuracy | plain vs inv_N [accuracy] | 10 | 16.0000 | 0.4688 | 0.0050 |
| accuracy | inv_p vs inv_N [accuracy] | 10 | 17.0000 | 0.5703 | 0.0050 |
| accuracy | PHPFL vs Standalone [linear] | 10 | 9.0000 | 0.0645 | 0.0325 |
| accuracy | PHPFL vs Standalone [entropy_correlated] | 10 | 0.0000 | 0.0020 | 0.0425 |
| accuracy | linear vs entropy [accuracy] | 10 | 15.0000 | 0.4102 | -0.0125 |
| \|acc - group mean\| | linear vs entropy [dispersion] | 10 | 7.5000 | 0.0410 | 0.0225 |
| PHPFL - Standalone | linear vs entropy [advantage] | 10 | 15.0000 | 0.4102 | 0.0125 |

## C smoke tests

| metric | comparison | n | stat | p | median diff |
|---|---|---|---|---|---|
| accuracy | PHPFL vs Standalone [linear] | 10 | 9.0000 | 0.0645 | 0.0325 |
| accuracy | PHPFL vs Standalone [entropy_correlated] | 10 | 0.0000 | 0.0020 | 0.0425 |
| accuracy | linear vs entropy [accuracy] | 10 | 15.0000 | 0.4102 | -0.0125 |
| \|acc - group mean\| | linear vs entropy [dispersion] | 10 | 7.5000 | 0.0410 | 0.0225 |
| PHPFL - Standalone | linear vs entropy [advantage] | 10 | 15.0000 | 0.4102 | 0.0125 |

# Appendix: per-client accuracies

## Five half-scale runs

| run | client | arch | acc | rare acc |
|---|---|---|---|---|
| PHPFL/lin | 0 | googlenet | 0.5800 | 0.0000 |
| PHPFL/lin | 1 | densenet121 | 0.6000 | 0.0000 |
| PHPFL/lin | 2 | efficientnet-b1 | 0.6760 | 0.0000 |
| PHPFL/lin | 3 | resnet18 | 0.7020 | 0.0000 |
| PHPFL/lin | 4 | googlenet | 0.6040 | |
| PHPFL/lin | 5 | densenet121 | 0.7100 | 0.0000 |
| PHPFL/lin | 6 | efficientnet-b1 | 0.5288 | |
| PHPFL/lin | 7 | resnet18 | 0.7360 | 0.0000 |
| PHPFL/lin | 8 | googlenet | 0.6880 | 0.0000 |
| PHPFL/lin | 9 | densenet121 | 0.4920 | 0.0000 |
| PHPFL/ent | 0 | googlenet | 0.5700 | 0.0000 |
| PHPFL/ent | 1 | densenet121 | 0.5600 | 0.0000 |
| PHPFL/ent | 2 | efficientnet-b1 | 0.6720 | 0.0000 |
| PHPFL/ent | 3 | resnet18 | 0.7100 | 0.0000 |
| PHPFL/ent | 4 | googlenet | 0.6080 | |
| PHPFL/ent | 5 | densenet121 | 0.6740 | 0.0000 |
| PHPFL/ent | 6 | efficientnet-b1 | 0.5329 | |
| PHPFL/ent | 7 | resnet18 | 0.7640 | 0.0000 |
| PHPFL/ent | 8 | googlenet | 0.6800 | 0.0000 |
| PHPFL/ent | 9 | densenet121 | 0.5340 | 0.1429 |
| Standalone/lin | 0 | googlenet | 0.5240 | 0.0000 |
| Standalone/lin | 1 | densenet121 | 0.4860 | 0.0000 |
| Standalone/lin | 2 | efficientnet-b1 | 0.5900 | 0.0000 |
| Standalone/lin | 3 | resnet18 | 0.6100 | 0.0000 |
| Standalone/lin | 4 | googlenet | 0.5400 | |
| Standalone/lin | 5 | densenet121 | 0.5620 | 0.0000 |
| Standalone/lin | 6 | efficientnet-b1 | 0.3683 | |
| Standalone/lin | 7 | resnet18 | 0.6900 | 0.0000 |
| Standalone/lin | 8 | googlenet | 0.6480 | 0.0000 |
| Standalone/lin | 9 | densenet121 | 0.4060 | 0.0000 |
| Standalone/ent | 0 | googlenet | 0.5260 | 0.0000 |
| Standalone/ent | 1 | densenet121 | 0.4620 | 0.0000 |
| Standalone/ent | 2 | efficientnet-b1 | 0.5960 | 0.0000 |
| Standalone/ent | 3 | resnet18 | 0.6400 | 0.0000 |
| Standalone/ent | 4 | googlenet | 0.5380 | |
| Standalone/ent | 5 | densenet121 | 0.5660 | 0.0000 |
| Standalone/ent | 6 | efficientnet-b1 | 0.3992 | |
| Standalone/ent | 7 | resnet18 | 0.6900 | 0.0000 |
| Standalone/ent | 8 | googlenet | 0.6540 | 0.0000 |
| Standalone/ent | 9 | densenet121 | 0.3920 | 0.0000 |
| FedAPEN/lin | 0 | googlenet | 0.5900 | 0.0000 |
| FedAPEN/lin | 1 | densenet121 | 0.5460 | 0.0000 |
| FedAPEN/lin | 2 | efficientnet-b1 | 0.6720 | 0.0000 |
| FedAPEN/lin | 3 | resnet18 | 0.6340 | 0.0000 |
| FedAPEN/lin | 4 | googlenet | 0.6120 | |
| FedAPEN/lin | 5 | densenet121 | 0.6660 | 0.0000 |
| FedAPEN/lin | 6 | efficientnet-b1 | 0.5021 | |
| FedAPEN/lin | 7 | resnet18 | 0.7900 | 0.0000 |
| FedAPEN/lin | 8 | googlenet | 0.6840 | 0.0000 |
| FedAPEN/lin | 9 | densenet121 | 0.5100 | 0.5000 |

## Diagnostic ladder runs

| run | client | arch | acc | rare acc |
|---|---|---|---|---|
| STEP1 | 0 | googlenet | 0.5460 | 0.3333 |
| STEP1 | 1 | densenet121 | 0.5660 | 0.0000 |
| STEP1 | 2 | efficientnet-b1 | 0.6120 | 0.0000 |
| STEP1 | 3 | resnet18 | 0.6980 | 0.0000 |
| STEP1 | 4 | googlenet | 0.5760 | |
| STEP1 | 5 | densenet121 | 0.6760 | 0.0000 |
| STEP1 | 6 | efficientnet-b1 | 0.5432 | |
| STEP1 | 7 | resnet18 | 0.7560 | 0.0000 |
| STEP1 | 8 | googlenet | 0.6700 | 0.0000 |
| STEP1 | 9 | densenet121 | 0.4540 | 0.2857 |
| STEP2 | 0 | googlenet | 0.5660 | 0.0000 |
| STEP2 | 1 | densenet121 | 0.5900 | 0.0000 |
| STEP2 | 2 | efficientnet-b1 | 0.6960 | 0.0000 |
| STEP2 | 3 | resnet18 | 0.7260 | 0.0000 |
| STEP2 | 4 | googlenet | 0.6220 | |
| STEP2 | 5 | densenet121 | 0.7240 | 0.0000 |
| STEP2 | 6 | efficientnet-b1 | 0.5370 | |
| STEP2 | 7 | resnet18 | 0.7380 | 0.0000 |
| STEP2 | 8 | googlenet | 0.6860 | 0.0000 |
| STEP2 | 9 | densenet121 | 0.4580 | 0.2143 |
| STEP3 | 0 | googlenet | 0.5740 | 0.0000 |
| STEP3 | 1 | densenet121 | 0.5780 | 0.0000 |
| STEP3 | 2 | efficientnet-b1 | 0.6800 | 0.0000 |
| STEP3 | 3 | resnet18 | 0.7320 | 0.0000 |
| STEP3 | 4 | googlenet | 0.5920 | |
| STEP3 | 5 | densenet121 | 0.7480 | 0.0000 |
| STEP3 | 6 | efficientnet-b1 | 0.5062 | |
| STEP3 | 7 | resnet18 | 0.7460 | 0.0000 |
| STEP3 | 8 | googlenet | 0.7020 | 0.0000 |
| STEP3 | 9 | densenet121 | 0.4500 | 0.0000 |
| STEP4 | 0 | googlenet | 0.5580 | 0.0000 |
| STEP4 | 1 | densenet121 | 0.5900 | 0.0000 |
| STEP4 | 2 | efficientnet-b1 | 0.6940 | 0.0000 |
| STEP4 | 3 | resnet18 | 0.7160 | 0.0000 |
| STEP4 | 4 | googlenet | 0.5960 | |
| STEP4 | 5 | densenet121 | 0.7480 | 0.0000 |
| STEP4 | 6 | efficientnet-b1 | 0.5288 | |
| STEP4 | 7 | resnet18 | 0.7500 | 0.0000 |
| STEP4 | 8 | googlenet | 0.6780 | 0.0000 |
| STEP4 | 9 | densenet121 | 0.4600 | 0.0000 |

Full client x class matrices are in phofl/analysis/matrices and two_axis_client_class.csv.

# Result artifact index

## Committed result files

| Path | Contents |
|---|---|
| phofl/results/A_fairness.csv | A fairness metrics |
| phofl/results/B_results.csv | B per-client aggregation results |
| phofl/results/C_results.csv | C per-client participation results |
| phofl/results/STATS_tests.csv | A/B/C Wilcoxon tests |
| phofl/results/C_holm.csv | C raw and Holm p-values |
| phofl/analysis/two_axis_run_summary.csv | Five-run two-axis summary |
| phofl/analysis/two_axis_fairness.csv | 40-row client/class table |
| phofl/analysis/two_axis_client_class.csv | 500-row client-class recall + support |
| phofl/analysis/two_axis_class_recall.csv | Global per-class recall |
| phofl/analysis/ranking_disagreement.csv | FM / class-gap ranks |
| phofl/analysis/am_normalization.csv | AM-normalized gaps |
| phofl/analysis/low_support_cells.csv | 170 excluded cells |
| phofl_half/results/C_half_summary.csv | Five-run summary |
| phofl_half/results/C_half_stats_full.csv | Half-scale tests + Holm |
| phofl_half/results/C_half_results.csv | 50 per-client rows |
| phofl_method/STEP*_summary.csv | Ladder per-step summary |
| phofl_method/STEP*_per_class.csv | Ladder per-class deltas |
| phofl_method/LADDER_SUMMARY.md | Ladder negative result |

Excluded: checkpoints, run_store, generated data, third_party, phofl_pinned, .venv-pinned. See DATA.md.

No method contribution is claimed. The participation hypothesis is closed. class_FM is a reporting metric, not a training objective.